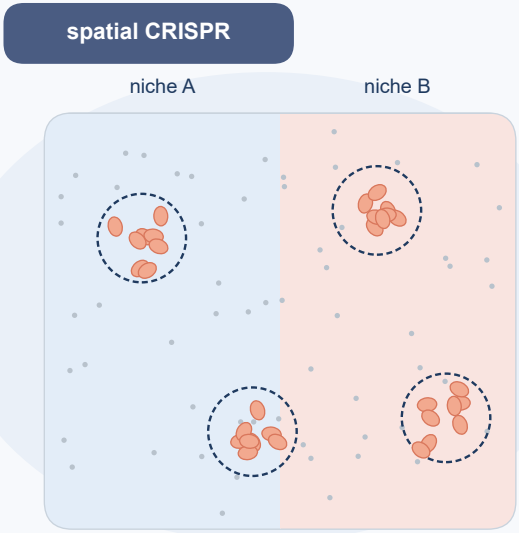
Does a knockout act differently in different tissue niches?



count cells

141 cells → p = 0.003

apparent niche effect

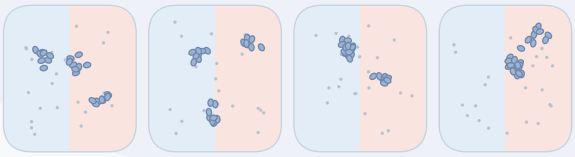
same knockout (CIITA, Xenium)

count lesions

10 lesions → p = 0.43

the lesion is the replicate

Before trusting a niche effect, calibrate it in the same tissue



control pseudo-knockouts with the same lesion sizes



rank the knockout's p among pseudo-knockouts

report Ψ ± CI

flag low power

62%

of null draws called niche-dependent by cell-level tests

20 lesions

per knockout: 83% power for a moderate niche effect

0.022 → 0.060

Tgfbr2 p before vs after calibration (Perturb-map)